COMP4141 Slides Week 9

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Admin

My Experience

MyExperience is open for you to leave your feedback so that this course can improve in future iterations. Please fill this out!

Due Dates

Due date for the week 8 homework has been moved from Monday, April 13 back to Friday, April 17. The final homework for week 9 will similarly be due on the Friday of week 10.

Exam Dates

The exam is 9 th of may.

In class assessment this week

Only 15 minutes this week.

coNP

coNP

coNP

coNP is the set of all languages $\Sigma^* \setminus A$, where A is a language in NP.

$P \subseteq (\text{coNP} \cap \text{NP})$

coNP-completeness

coNP-completeness

A language L is coNP-complete if

- $L \in$ coNP, and
- every problem in coNP is polynomial-time reducible to L.

It is unknown whether NP = coNP.

PSPACE

PSPACE and NPSPACE

PSPACE and NPSPACE

Let $f : \mathbb{N} \to \mathbb{N}$. Define the space complexity classes $\mathrm{SPACE}(f(n))$ and $\mathrm{NSPACE}(f(n))$ as the collections of all languages decidable by a TM using $O(f(n))$ space and an NTM using $O(f(n))$ space, respectively.

Example: SAT

SAT Space Complexity

$\mathrm{SAT} \in \mathrm{SPACE}(n)$

PSPACE and NPSPACE

PSPACE $= \bigcup_{k \in \mathbb{N}} \mathrm{SPACE}(n^k)$ and NPSPACE $= \bigcup_{k \in \mathbb{N}} \mathrm{NSPACE}(n^k)$.

SPACE vs. NSPACE

Savitch's Theorem

For any function $f : \mathbb{N} \to \mathbb{N}$ with $f(n) \geq n$,

$$\mathrm{NSPACE}(f(n)) \subseteq \mathrm{SPACE}(f(n)^2).$$

Corollary

PSPACE = NPSPACE.

PSPACE-Completeness

PSPACE-Completeness

A language B is PSPACE-complete if

1. $B \in$ PSPACE, and
2. B is PSPACE-hard, i.e., every $A \in$ PSPACE is polynomial-time reducible to B.

Quantified Boolean Formulas

Quantified Boolean Formulas

$$\varphi \to p \mid \varphi \wedge \varphi \mid \neg\varphi \mid (\varphi) \mid \forall p(\varphi) \mid \exists p(\varphi)$$

$$\forall x(\varphi) = \varphi[\text{true}/x] \wedge \varphi[\text{false}/x], \quad \exists x(\varphi) = \varphi[\text{true}/x] \vee \varphi[\text{false}/x]$$

$$\text{QBF} = \{\langle\varphi\rangle \mid \varphi \text{ is true}\}$$

QBF Space Complexity

QBF is PSPACE-complete.

Two-Player Games and PSPACE-Completeness

Complexity of Two-Player Games

- Let ψ be a Boolean formula over propositions $x_1, \ldots, x_k$. In move i, a player assigns a truth value to x_i. The problem

$$\{\langle\psi\rangle \mid E \text{ has a winning strategy}\}$$

  is **PSPACE**-complete.
- Geography Game is **PSPACE**-complete.
- Deciding whether a given $n \times n$ Go position is winning is **PSPACE**-hard.

The question: "Does player E have a winning strategy from position p in game G?" can often be expressed as a quantified Boolean formula:

$$\exists m_1 \forall r_1 \exists m_2 \cdots \varphi.$$

L and NL

L and NL

Definition

$\mathsf{L} = \mathsf{SPACE}(\log n), \quad \mathsf{NL} = \mathsf{NSPACE}(\log n).$

Example of NL

s-t Reachability

Input: A directed graph G and vertices s, t

Question: Is t reachable from s in G?

Log-Space Reducibility

Definition

A log-space transducer is a 3-tape TM with:

- a read-only input tape,
- a read-write work tape, and
- a write-only output tape,

that uses only $O(\log n)$ space on the work tape.

We use log-space reducibility ($\leq_L$) based on such transducers.

NL-completeness

NL-completeness

A language B is NL-complete if $B \in \mathrm{NL}$ and for every $A \in \mathrm{NL}$ we have $A \leq_L B$.

Theorem

s-t Reachability is NL-complete.

coNL and NL

coNL

coNL is the set of all languages $\Sigma^* \setminus A$, where A is a language in NL.

Theorem (Immerman–Szelepcsényi)

NL = coNL.

Proof idea: show that the complement of s-t Reachability is in NL.

W9 P1

Problem 1

Show that if L is an **NP**-complete language, then $\overline{L}$ is **coNP**-complete.

W9 P2

Problem 2

Show that the following problem is **coNP**-complete:

Validity

Input: A Boolean formula φ

Question: Is φ true for all truth assignments?

W9 P3

Problem 3

A ladder in a language L is a sequence of strings in L, all of the same length, such that each differs from the preceding one by at most one character. For example, the following is a ladder in English, starting with "head" and ending in "free".

head, hear, near, fear, bear, beer, deer, deed, feed, feet, fret, free.

Let $LADDER_{DFA} = \{\langle M, s, t\rangle \mid M \text{ is a DFA and } L(M) \text{ contains a ladder from } s \text{ to } t\}$. Show that $LADDER_{DFA} \in$ **PSPACE**.

W9 P4

Problem 4

Show that the following problem is **PSPACE**-complete:

Linear-bounded DTM acceptance (LBDA)

Input: A Turing Machine M and a word w and a natural number k (in unary)

Question: Does M accept w using at most $k|w|$ tape cells?

W9 P5

Problem 5

Consider the following decision problem

2-SAT

Input: A boolean formula φ in 2-CNF (at most 2 literals per clause

Question: Is φ satisfiable?

(a) Show that 2-SAT$\in$ P

Problem 5

Consider the following decision problem

2-SAT

Input: A boolean formula φ in 2-CNF (at most 2 literals per clause

Question: Is φ satisfiable?

(b) Show that 2-SAT$\in$ NL

W8 P5

Problem 5

Consider the following problem:

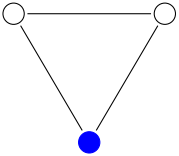
3-Col

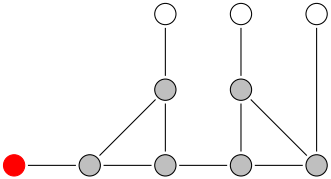
Input: An undirected graph G

Question: Is there a colouring of the vertices of G using at most 3 colours such that no edge has both endpoints with the same colour?

Problem 5

(a) What can be said about the colouring of the white vertices in the following (partially coloured) graphs?





Problem 5

(b) Show that 3-Col is **NP**-complete, assuming that SAT is **NP**-complete.

W8 P6

Problem 6

Show that if $\mathbf{P} = \mathbf{NP}$, then every language $A \in \mathbf{P}$ except $A = \emptyset$ and $A = \Sigma^*$, is $\mathbf{NP}$-complete.